

Atishaya Maharjan

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EDUCATION

BSc. Joint Honours: Computer Science & Mathematics, Minor in Statistics Aug. 2022 – Present
University of Manitoba *Winnipeg, Canada*

International Undergraduate Student Scholarship | Dean's Honour Roll

PROFESSIONAL EXPERIENCE

Software Engineer Mar. 2024 – Present
LG Electronics *Englewood Cliffs, USA (Remote)*

- Built features for a B2B platform serving **1,000+** vendors using **Next.js**, **Redux**, and **TypeScript**, contributing to a **25%** increase in user engagement.
- Built analytics and custom tagging frameworks used by **6+** product teams to support product measurement and decision-making.
- Developed global-navigation AI search that reduced average latency by **35 ms**; implemented promo-code and magic-link workflows associated with a **15%** increase in conversion.

Software Engineer
Vibecore – Cloud Infrastructure & CI/CD

- Automated CI/CD workflows from **GitHub Actions** to **AWS EC2** instances for application deployment.
- Managed **S3** buckets, **DynamoDB** integrations, and **IAM** permissions for cloud-connected application workflows.
- Provisioned and maintained **EBS/EFS** storage for EC2-hosted services and deployment infrastructure.

Software Engineer Jun. 2025 – Jan. 2026
Hoek Insights *Amsterdam, Netherlands (Remote)*

- Built a **Python** package embedding **Three.js** visualizations in **Jupyter Notebooks** via **React** renderers, cutting render time by **38%** and bundle size by **25%**.
- Designed a **FastAPI/PostgreSQL** MVP with **18+** endpoints and **10** workflows for a marketing platform.
- Built integration-test coverage to **95%** and maintained below **200 ms** median API latency in staging.

Software Engineer Aug. 2019 – Apr. 2024
Timeero *Phoenix, USA (Remote)*

- Led frontend development of a mileage-tracking application using **TypeScript**, **React**, **Vite**, and **Mapbox**.
- Supported production workflows for **1,000+** active users across mileage tracking and time-capture features.
- Implemented **15+** geospatial segment-mapping features, improving time-capture accuracy by **30%**.

SELECTED PROJECTS

Brainstorm 🔗 | *Full-Stack, RAG, Data Systems*

- Built a role-aware knowledge-management platform integrating **Jira**, **PostgreSQL**, **Redis**, and **Pinecone** through retrieval-augmented generation.
- Designed role-based retrieval and source grounding across **4** connected systems, restricting responses to authorized organizational data.
- Implemented multi-model routing with dynamic selection and fallback policies balancing latency, reliability, and response quality.

WinLocate 🔗 | *Full-Stack, Open Data, AI*

- Built a full-stack data platform using **Socrata** open data for companies considering relocation to Winnipeg.
- Implemented city comparisons, zoning maps, budget tools, and relocation-planning workflows for applied civic data analysis.
- Won **1st place** among **15** teams at the Winnipeg Open Data Datathon 2026.

Get Good Grades 🔗 | *Electron, React, TypeScript, YAML*

- Built a decentralized flashcard viewer that ingests **.yaml** flashcard files from anywhere in the file system.
- Gave users full control over flashcard content and scheduling logic instead of relying on centralized storage.
- Presented the project at DevHacks 2026.

Personal Portfolio Website | *Astro, React*

- Built a terminal-style server-side rendered portfolio with an interactive command spotlight feature and **15+** custom commands.
- Designed modular command routing, theming, and reusable **React** components across **6+** content sections.
- Implemented the portfolio with **Astro** and **React** to keep content sections reusable and maintainable.

SELECTED RESEARCH EXPERIENCE

Algorithms & Computational Geometry | *GADA Lab* Mar. 2024 – Present

- Developing faster exact and approximation algorithms for fuel-efficient collaborative delivery using dynamic programming, exchange-point geometry, and structural properties of optimal delivery sequences.
- Developed an approximation algorithm for Hamiltonian-path optimization in spatially embedded graphs , with visualization and randomized testing infrastructure for **100+** synthetic instances.
- Improved the performance of a noncrossing spanning-tree research library by **30%** and introduced stronger **Python** development practices.

Machine Learning Research | *Terrabyte Machine Learning Research Group* May 2025 – Present

- Generalized Graph Neural Diffusion architectures to hypergraphs using **PyTorch** and **PyTorch Geometric**.
- Implemented **4** Hypergraph Neural Diffusion variants for graph and hypergraph machine-learning experiments.
- Built a reproducible **MLflow** pipeline tracking **25+** experiments across **5+** datasets.

Combinatorics & Graph Theory | *University of Manitoba* Jan. 2024 – Jul. 2025

- Established a coefficientwise inequality underlying the Hurlbert–Kamat property for perfect m -ary trees.
- Derived a closed-form expression for the independence number of perfect m -ary trees using generating functions and recurrence analysis.
- Used induction and **SageMath** validation to test conjectures and verify symbolic identities across **1,000+** instances.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C++, Java, SQL, Bash

Frameworks & Libraries: React, Next.js, Redux, FastAPI, Node.js, PyTorch, Three.js

Data: PostgreSQL, Redis, MongoDB, DynamoDB, Pinecone

Cloud & Infrastructure: AWS EC2, S3, IAM, EBS/EFS, Docker, GitHub Actions, Linux

Tools & Methods: Git, Jupyter, RAG, Model Context Protocol, Mapbox

SELECTED AWARDS

CSSA Capture the Flag – 1st of 41 teams (2026), 2nd of 25+ teams (2025)

Winnipeg Open Data Datathon – 1st Place among 15 teams 2026

S.U.R.E. Poster Competition – 3rd Place, Computational Science Aug. 2024

- Recognized for the research poster “*Minimizing Maximum Interference in 1-Dimensional Networks*” at the University of Manitoba.

.devHacks – Most Chaotic Evil Hack among 20+ teams

LEADERSHIP

CS Help Center Leader & Teaching Assistant | *University of Manitoba* Jan. 2025 – Apr. 2025

- Mentored **100+** students across first- through third-year CS courses and graded **150+** assignments for Analysis of Algorithms (COMP 3170).

Events Executive | *.devClub (University of Manitoba)* Aug. 2023 – Jul. 2024

- Co-led workshops, hackathons, academic review sessions, and community events for the Computer Science Department.

Coding Kickoff Facilitator | *University of Manitoba* Feb. 2025 – Mar. 2025

- Facilitated workshops teaching **IDE setup** and **Git** to **50+** first-year university students.

Audio Visual Team Member | *Bethesda Church* Jul. 2023 – Present

- Operated A/V equipment for weekly sermons and live events, ensuring high-quality audiovisual experiences for attendees.